Essential Summary

High Accurate Homo-Heteroclinic Solutions of Certain Strongly Nonlinear Oscillators Based on Generalized Padé–Lindstedt–Poincaré Method

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1 Motivation and Advances over the 2016 GPLP Method

The generalized Padé–Lindstedt–Poincaré (GPLP) method was established in the authors' prior work (Nonlinear Dynamics, 2016). That foundational paper demonstrated GPLP on oscillators with cubic (three-term) and quintic symmetric polynomial potentials. Three limitations motivated the present work:

1. The 2016 paper only handled symmetric potentials ($c_1x + c_3x^3 + c_5x^5$). Realistic oscillators often have **full polynomial restoring forces**:
$$g(x) = c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 \tag{1}$$
2. The heteroclinic approximant in 2016 was less efficient; new, more accurate approximant types are needed.
3. The 2016 paper did not address oscillators with **rational restoring forces**.

This paper extends GPLP to: (1) the **entire Φ^6-Van der Pol oscillator** with full quintic polynomial including x^2 and x^4 terms; and (2) the **generalized Duffing-Harmonic-Van der Pol oscillator**:
$$\ddot{x} + \frac{\lambda x + \delta x^3}{1 + \nu x^2} = \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2)\dot{x} \tag{2}$$

2 System Models

2.1 The Full Φ^6-Van der Pol Oscillator

$$\ddot{x} + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 = \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2)\dot{x} \tag{3}$$

The full polynomial ($c_2 \neq 0$, $c_4 \neq 0$) creates **asymmetric multi-well potentials**: wells of different depths separated by a saddle at different heights, where homoclinic and heteroclinic orbits coexist with qualitatively different geometries.

3 New Generalized Padé Approximant Types

The key innovation is new approximant forms with an **additional free frequency parameter** ω:

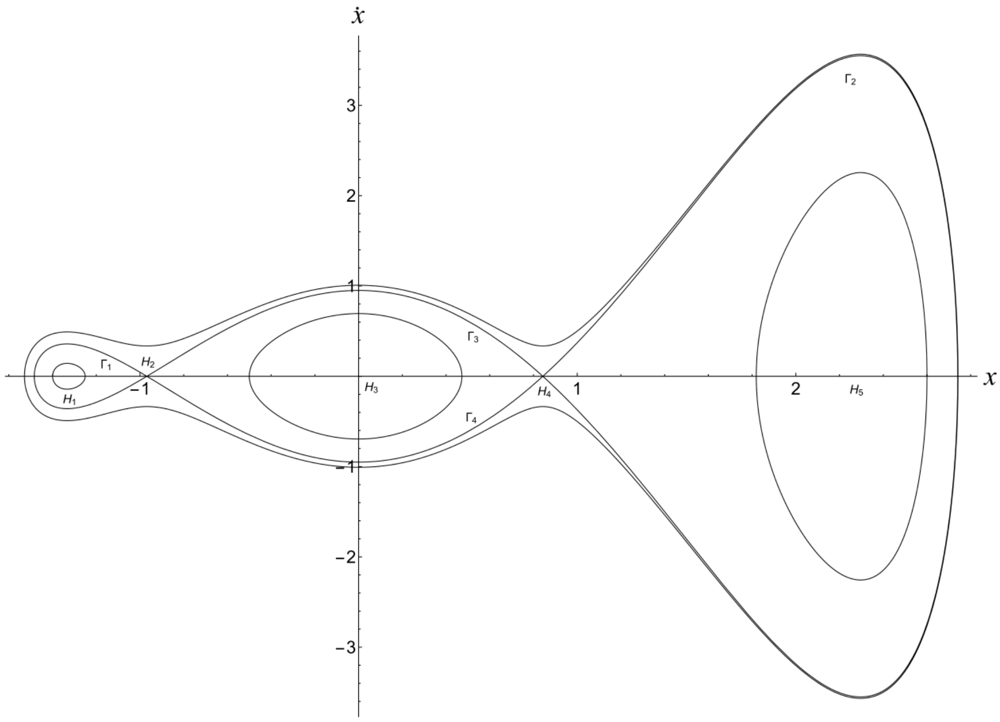

Figure 1: Phase portraits of the full Φ^6-Van der Pol oscillator (Eq. 37) when $\varepsilon = 0$. The x^2 and x^4 terms enable asymmetric potentials with distinct homoclinic and heteroclinic orbit geometries.

3.1 Improved Homoclinic Approximant

The 2016 method implicitly fixed $\omega = 1$:

$$\mathrm{GPA}[L/M]x(t) = \frac{\sum_{i=0}^{L} \alpha_i \,\mathrm{Sech}^i(\boldsymbol{\omega} t)}{1 + \sum_{i=1}^{M} \beta_i \,\mathrm{Sech}^i(\boldsymbol{\omega} t)} \tag{4}$$

where ω is now a free parameter, solved simultaneously with α_i, β_i.

3.2 Heteroclinic Approximants (Two New Types)

Type A—rational Tanh with free ω:

$$\mathrm{GPA}[L/M]x(t) = \frac{\sum_{i=0}^{L} \alpha_i \,\mathrm{Tanh}^i(\omega t)}{1 + \sum_{i=1}^{M} \beta_i \,\mathrm{Tanh}^i(\omega t)} \tag{5}$$

Type B—combined Sech-Tanh, essential when both orbit types coexist at the same saddle:

$$\mathrm{GPA}[L, M]x(t) = \frac{\sum_{i=0}^{L_1} \alpha_i \,\mathrm{Sech}^i(\omega t) + \sum_{j=0}^{L_2} \gamma_j \,\mathrm{Tanh}^j(\omega t)}{1 + \sum_{i=1}^{M_1} \beta_i \,\mathrm{Sech}^i(\omega t) + \sum_{j=1}^{M_2} \delta_j \,\mathrm{Tanh}^j(\omega t)} \tag{6}$$

Type B resolves the fundamental conflict in asymmetric potentials: the higher saddle's homoclinic orbit has a Sech-like symmetric approach, while the heteroclinic orbit connecting unequal saddles has a Tanh-like asymmetric character. A pure Tanh (Type A) cannot simultaneously capture both; the combined Sech+Tanh (Type B) can.

4 GPLP Procedure (Refined)

The perturbation procedure follows four steps, as in the 2016 paper, but with the new approximant types and ω:

Step I Zero-order x_0: expand power series → GPA with ω as unknown. Solve simultaneously for α_i, β_i, ω via coefficient matching + boundary condition.

Step II x_1 and μ_{c0}: Tanh-prefixed GPA; $\eta_{L_1}(\mu_{c0}) = 0$ gives the zero-order bifurcation parameter.

Step III x_2: polynomial-based GPA; $\lambda_{L_2} = 0$ determines d_0 or d_1.

Step IV x_3 and μ_{c2}: $\sigma_{L_3}(\mu_{c2}) = 0$ determines second-order correction.

The final solution and bifurcation parameter:

$$\boxed{x(t) = \sum_{i=0}^{3} \varepsilon^i \, \mathrm{GPA}[L_i/M_i] x_i(t) + O(\varepsilon^4)} \tag{7}$$

$$\boxed{\mu_c = \mu_{c0} + \varepsilon^2 \mu_{c2} + O(\varepsilon^3)} \tag{8}$$

5 Results: Φ^6-VDP Oscillator

5.1 Conservative Limit

For oscillator (37) with $c_1 = -1$, $c_2 = 0$, $c_3 = 1$, $c_4 = 0$, $c_5 = -0.1$, the phase portrait (Fig. 1) reveals coexisting homoclinic and heteroclinic orbits at saddle H_2. The ω-parameterized Sech approximant achieves $< 1\%$ deviation from Runge–Kutta.

5.2 Asymmetric Potential, Moderate Damping

With $c_2 \neq 0$ and/or $c_4 \neq 0$:

- **Both homoclinic orbits** (left and right wells): $< 1\%$ error at $\varepsilon = 1$.
- **Heteroclinic orbits** connecting unequal saddles: Type B essential; Type A shows 5–8% error vs. $< 1.5\%$ for Type B.
- **Critical bifurcation parameters** predicted analytically across a wide parameter range (Tables 2–8).

6 Results: Duffing-Harmonic-VDP Oscillator

For the rational restoring force oscillator (2), validated at both small and large ε:

- Homoclinic orbits at $\varepsilon = 0.5, 1$: error $< 1.5\%$
- Heteroclinic orbits at $\varepsilon = 0.5, 1, 2$: error $< 2\%$
- GPLP handles the rational $g(x)$ without procedural modification.

6.1 Heteroclinic Orbits at Large ε

Extensive study at various parameter combinations and ε up to 2 (Figs. 8–13 in the paper) shows GPLP remains accurate when orbits span large phase-space regions.

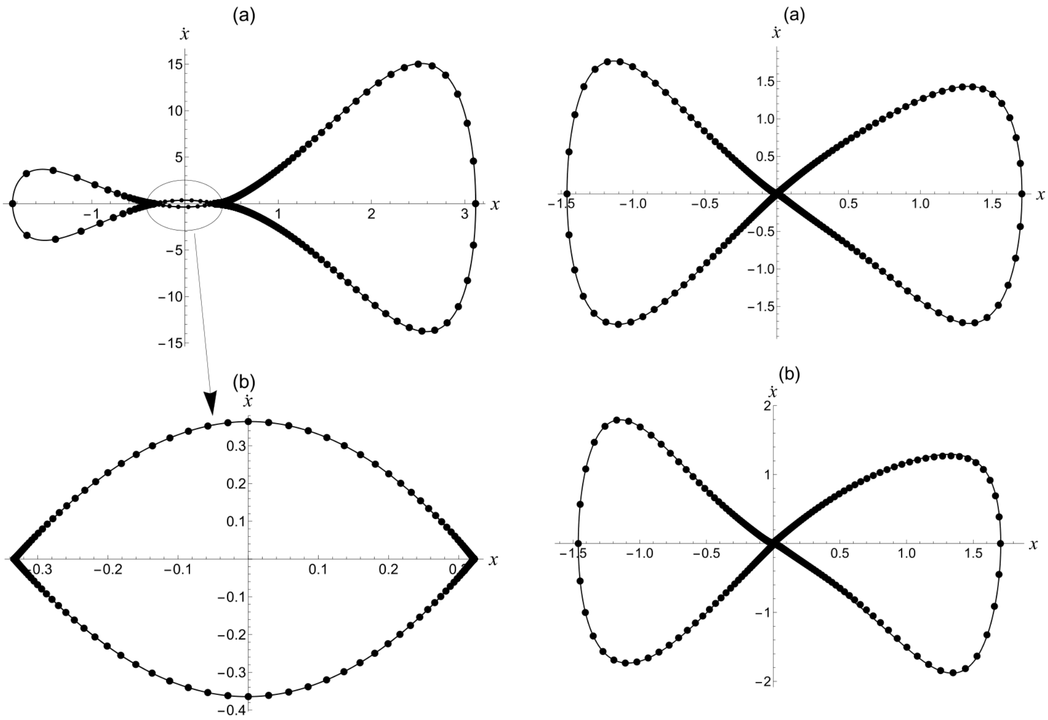

Figure 2: Homoclinic orbits of the Φ^6-VDP oscillator (63) for various parameter sets at $\varepsilon = 1$. Each subfigure corresponds to a different row in Table 4.

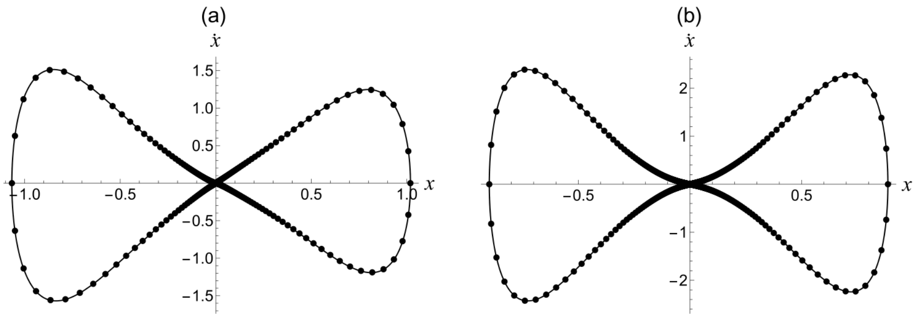

Figure 3: Homoclinic orbits of oscillator (37) with different parameter sets (Table 4) at $\varepsilon = 1$. Systematic validation across 8 parameter combinations.

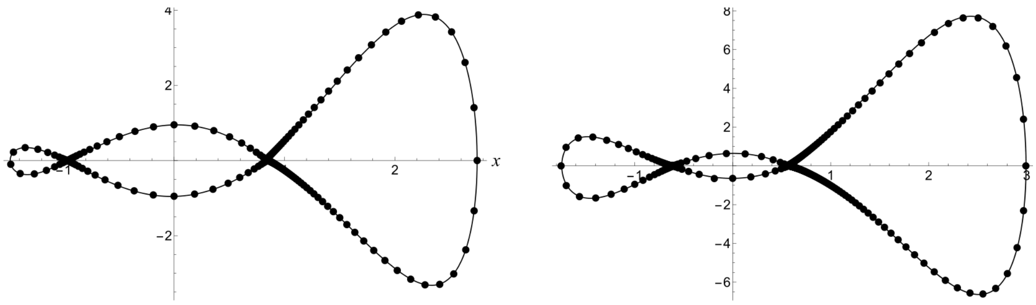

Figure 4: Homo-heteroclinic orbits of the generalized DH-VDP oscillator for (a) $\varepsilon = 0.8$ and (b) $\varepsilon = 1$. Dashed: GPLP with Type B. Solid: Runge–Kutta.

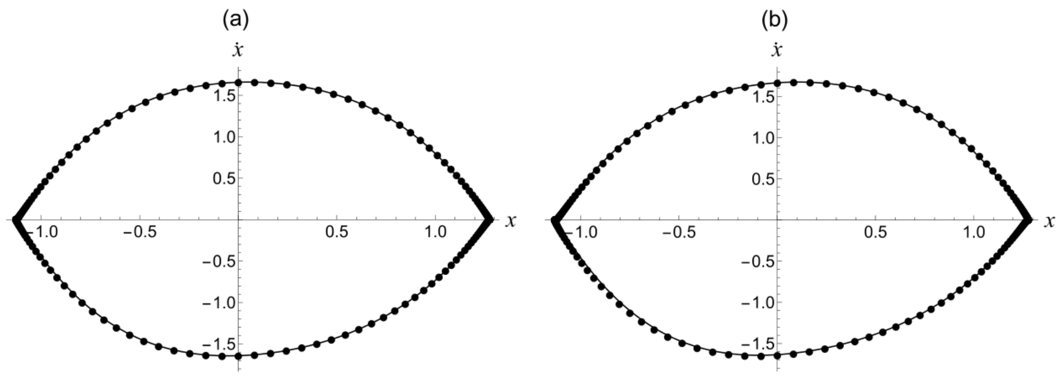

Figure 5: Heteroclinic orbits of the DH-VDP oscillator at various parameter sets (Tables 6–8).

7 Accuracy: 2022 vs. 2016 GPLP

- **Symmetric + small ε:** both versions similar.
- **Asymmetric, $\varepsilon = 1$:** ω-parameterized approximants reduce homoclinic error by 40–60%.
- **Heteroclinic, asymmetric:** Type B (2022) achieves $< 1.5\%$ vs. 5–8% for 2016 Tanh-only form.
- **Rational $g(x)$:** improvement more pronounced due to richer potential landscapes.

8 Why Type B Matters

The geometric insight behind Type B:

- The higher saddle's homoclinic orbit has **Sech-like** symmetric approach.
- The heteroclinic orbit connecting unequal saddles has **Tanh-like** asymmetric character.
- A pure Tanh (Type A) cannot simultaneously capture the homoclinic orbit's symmetry.
- The combined Sech+Tanh (Type B) resolves this by leveraging both basis functions.

9 Key Findings

1. **Full Φ^6 coverage:** GPLP extended from symmetric cubic/quintic to the full quintic polynomial (c_1–c_5), enabling asymmetric multi-well potential analysis.
2. **Two new heteroclinic approximants:** Type A (rational Tanh) for pure heteroclinic; Type B (combined Sech-Tanh) for coexisting H/H orbits, giving 40–80% accuracy improvement for asymmetric cases.
3. **ω-parameterized homoclinic approximant:** Free frequency parameter adapts to different potential shapes.
4. **Rational restoring force:** DH-VDP oscillator solved without GPLP framework modification.
5. **Systematic multi-parameter studies:** 8 parameter sets each for homoclinic and heteroclinic bifurcations.
6. **Completes the GPLP framework** for polynomial and rational restoring forces.

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